

Final Integrative Project

ACT AS A BUSINESS INTELLIGENCE TEAM · END-TO-END

Team size: 4 students

Grade: group grade (1.0–7.0)

Language: English (all deliverables)

Roster + topic due: Jul 19, 23:55

Package due: Aug 11, 23:55

Presentation: Aug 12, in class

Your team is a **Business Intelligence unit** hired by an organization. Starting from a real dataset and a real business question, you run the **full BI cycle** you have practised all semester and deliver both a working analysis and a decision your stakeholders can act on. You own the framing, the data, the model, the dashboard, and the recommendation. All four members contribute and all four present.

1 · The assignment

what you build

Choose a real-world dataset and a concrete business question, then take it end to end: frame the problem and its KPIs, prepare and understand the data, build and honestly evaluate a predictive or segmentation model, communicate the result through a dashboard, and close with an ethics and limitations review. The output is a reproducible notebook plus a stakeholder presentation, as if you were reporting to a client or an internal decision-maker.

FRAMING

Frame every choice around the business decision it serves, then the technique that supports it. State who the decision-maker is, what they will do differently because of your work, and how success is measured. A correct model that answers no business question scores low; a modest model tied to a clear, well-argued decision scores well.

2 · Teams and topic registration

by Jul 19

- Teams of **4 students**. If the class size does not divide evenly, teams of **3** are allowed; teams of 5 only as an exception authorized by the instructor.
- One member per team (the **team representative**) sends a single email with: **(a)** the full roster (names + student IDs of all members), and **(b)** the chosen topic: dataset, source link, and a one-paragraph business question.
- Deadline: **Sunday, July 19, 2026, 23:55**. Email subject: **BI2026 Final Project - Team [name] - Topic**.
- Two teams may use the same public dataset only if their business questions clearly differ. The instructor confirms each topic; if a topic is too narrow or too broad, you get one revision.

3 · Dataset requirements

choose well

You pick the dataset. It must satisfy every criterion below. A weak dataset caps the whole project.

REQUIRED

- Public and documented **in English** (has a data dictionary or column descriptions).
- At least **~1,000 rows** and **~6 columns**, mixing numeric and categorical fields.
- Supports a real task: a **target to predict** (regression/classification) **or** a meaningful **segmentation** problem.
- A genuine business context you can name (retail, finance, health ops, mobility, energy, etc.).

NOT ALLOWED

- Confidential, proprietary, or personally identifying data you are not authorized to use.
- Toy teaching datasets already used in class (Iris, Titanic, the built-in seaborn/sklearn sets).
- Datasets with no plausible decision attached to them.

GOOD SOURCES

UCI ML Repository, Kaggle Datasets, data.gov / datos.gob.cl and other open-government portals, World Bank, official company or sector open data.

4 · Required scope: the BI pipeline

all five stages

Your notebook must move through the five stages below. Each stage maps to a course unit and to the rubric.

STAGE 1 Frame & KPIs Business question, decision-maker, success metric and KPIs. Unit I	STAGE 2 Prepare data EDA, cleaning, missing/outliers, encoding, features. No leakage. Unit II	STAGE 3 Model & evaluate Prediction or segmentation, proper validation, honest metrics. Unit II-III	STAGE 4 Communicate Dashboard (Looker/Power BI or HTML) + a recommendation to act on. Unit IV	STAGE 5 Ethics & limits Bias, fairness, privacy, and honest limitations of your result. Unit IV
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TECHNICAL BASELINE

Python with `pandas`, `scikit-learn`, and `matplotlib` (the course stack). Validation must be sound. Use train/test or k-fold cross-validation, with metrics appropriate to the task (for example precision/recall/F1/ROC-AUC for classification, RMSE/MAE for regression, and a defensible choice of k plus a cluster-quality check for segmentation). Set a random seed so results are stable.

5 · What you submit

one ZIP

The team representative submits **one** ZIP file named `BI2026_FinalProject_Team-[name].zip`, containing:

Item	Requirement
Reproducible notebook .ipynb	Runs top to bottom without errors on a clean environment (Google Colab compatible). Loads the dataset from the included file or a documented public URL. Seeded. Markdown cells narrate each of the five stages. Any AI assistance is disclosed (see §7).
Presentation .pdf	The exact slide deck you present, exported to PDF. Built for a stakeholder audience: business framing first, honest results, one clear recommendation. The dashboard may be embedded screenshots, a working Looker Studio / Power BI link, or a self-contained <code>.html</code> dashboard included in the ZIP, as long as its data connections work and it reproduces from your dataset.
Dataset .csv / .parquet	The exact data you used. If it exceeds ~50 MB, include a representative sample plus a <code>DATA_SOURCE.md</code> with the full download link and instructions to reproduce.
README .md	How to run the notebook, the dataset source and license, and a contribution table listing each of the 4 members and what they did (this documents that everyone participated).

REPRODUCIBILITY CHECKLIST

- ☐ Fresh "Restart & Run all" produces every figure and number with no manual steps.
- ☐ Random seed set; no hidden cells; no absolute local paths.
- ☐ Every table and chart in the slides can be traced back to a notebook cell.

6 · The presentation

Aug 12, in class

- **15 minutes** of presentation + up to **5 minutes** of questions from the panel.
- **All four members must speak.** Distribute the narrative so each person owns part of it.

- Present to a stakeholder or client audience. Lead with the business question and the recommendation, and keep the technical depth honest but proportionate.
- Questions test whether you understand your own work: expect to justify a metric, a data choice, a limitation, or a line of code.
- Presentation order is announced in advance; attendance is mandatory for the whole session.

7 - Deadlines, AI policy, integrity

[read carefully](#)

REGISTER

Jul 19

Roster + topic, 23:55

SUBMIT ZIP

Aug 11

Full package, 23:55

PRESENT

Aug 12

In class, 15 min + Q&A

AI USE POLICY

AI tools (assistants, code helpers, NL-to-SQL) are **allowed as tools**. You must **disclose** their use in the README and you must **understand and defend every line** you submit. In Q&A, code you cannot explain counts against you. Undisclosed heavy reliance is treated as an integrity issue.

INTEGRITY & LATE WORK

The work must be your team's own. Cite any external code or ideas. Copying another team's analysis, or submitting results the notebook does not reproduce, voids the grade for that component. The ZIP deadline is a **hard close at 23:55**; late submissions follow UDD regulations.

8 • Grading rubric

group grade • 1.0–7.0

The project is scored over **100 points** across eight criteria. Each criterion is placed at one of four achievement levels, worth **100% / 80% / 55% / 25%** of its weight (the grader may interpolate within a level). The total is converted to a 1.0–7.0 grade with a **60% passing standard** (4.0 at 60 points). The grade is the **same for all team members**.

Criterion	Exemplary 100%	Proficient 80%	Developing 55%	Insufficient 25%
Business framing & KPIs 12 pts • Unit I	Sharp business question, named decision-maker, well-chosen KPIs and success metric tied to a real decision.	Clear question and KPIs; link to a decision is present but partly generic.	Question stated but vague; KPIs weak or loosely connected to the analysis.	No real business framing; the work is a technical exercise with no decision.
Data prep & quality 14 pts • Unit II	Thorough EDA; principled handling of missing values, outliers, encoding; features justified; leakage explicitly avoided.	Solid cleaning and features; minor gaps or an unjustified step.	Basic cleaning only; quality issues left unaddressed or a possible leak.	Little to no preparation; data used as-is with evident quality problems.
Modeling & evaluation 18 pts • Unit II–III	Model fits the question; validation is correct (hold-out / CV); metrics are appropriate and honestly reported with a baseline.	Reasonable model and validation; metric choice mostly right; baseline weak or missing.	Model runs but validation is flawed (e.g. test leakage) or metrics misread.	No valid evaluation; results not trustworthy or not interpretable.
Insight, viz & recommendation 14 pts • Unit IV	Analysis becomes a clear, actionable recommendation; visuals/dashboard are honest, well-chosen, and readable.	A recommendation is given and mostly supported; visuals are adequate.	Findings listed but no real decision; charts unclear or misleading in places.	No usable insight; visuals absent, wrong, or actively misleading.
Ethics, bias & limitations 10 pts • Unit IV	Concrete bias/fairness and privacy analysis; limitations named honestly with impact on the decision.	Real discussion of ethics and limits; somewhat generic or incomplete.	Token mention of ethics/limitations with no specificity.	Absent; overclaims results with no caveats.
Reproducibility & packaging 12 pts	“Restart & Run all” works cleanly; seeded; ZIP complete and well-structured; README lets anyone reproduce.	Runs with a trivial fix; package mostly complete.	Runs only with manual intervention; missing item or unclear instructions.	Does not reproduce; package incomplete.
Communication & presentation 14 pts	Compelling stakeholder narrative, strong English, within time; Q&A answered with command of the work.	Clear and on-time; English fluent enough; Q&A handled reasonably.	Disorganized or over/under time; language or Q&A struggles.	Unclear, far off time, or cannot answer basic questions.
Teamwork & participation 6 pts	All four contribute and present; contribution table credible; the work reads as one coherent team effort.	All participate; contribution slightly uneven.	Participation clearly unbalanced; parts feel disconnected.	One or more members do not present or contribute.

POINTS → GRADE (60% STANDARD)

```
p = points / 100
if p < 0.60 : N = 1.0 + 5.0·p
if p ≥ 0.60 : N = 4.0 + 7.5·(p - 0.60)
```

Linear on each side. 0 pts → 1.0, 60 pts → 4.0 (pass), 100 pts → 7.0. Rounded to one decimal.

CONVERSION TABLE

Points	Grade	Points	Grade
100	7.0	60	4.0
90	6.3	50	3.5
80	5.5	40	3.0
70	4.8	30	2.5
65	4.4	20	2.0
60	4.0	0	1.0

WORKED EXAMPLE

A team scores Proficient on framing (9.6), data prep (11.2), modeling (14.4), insight (11.2), reproducibility (9.6), communication (11.2), Exemplary on teamwork (6.0), and Developing on ethics (5.5). Total ≈ **78.7 points** → p = 0.787 → N = 4.0 + 7.5·(0.187) ≈ **5.4**.